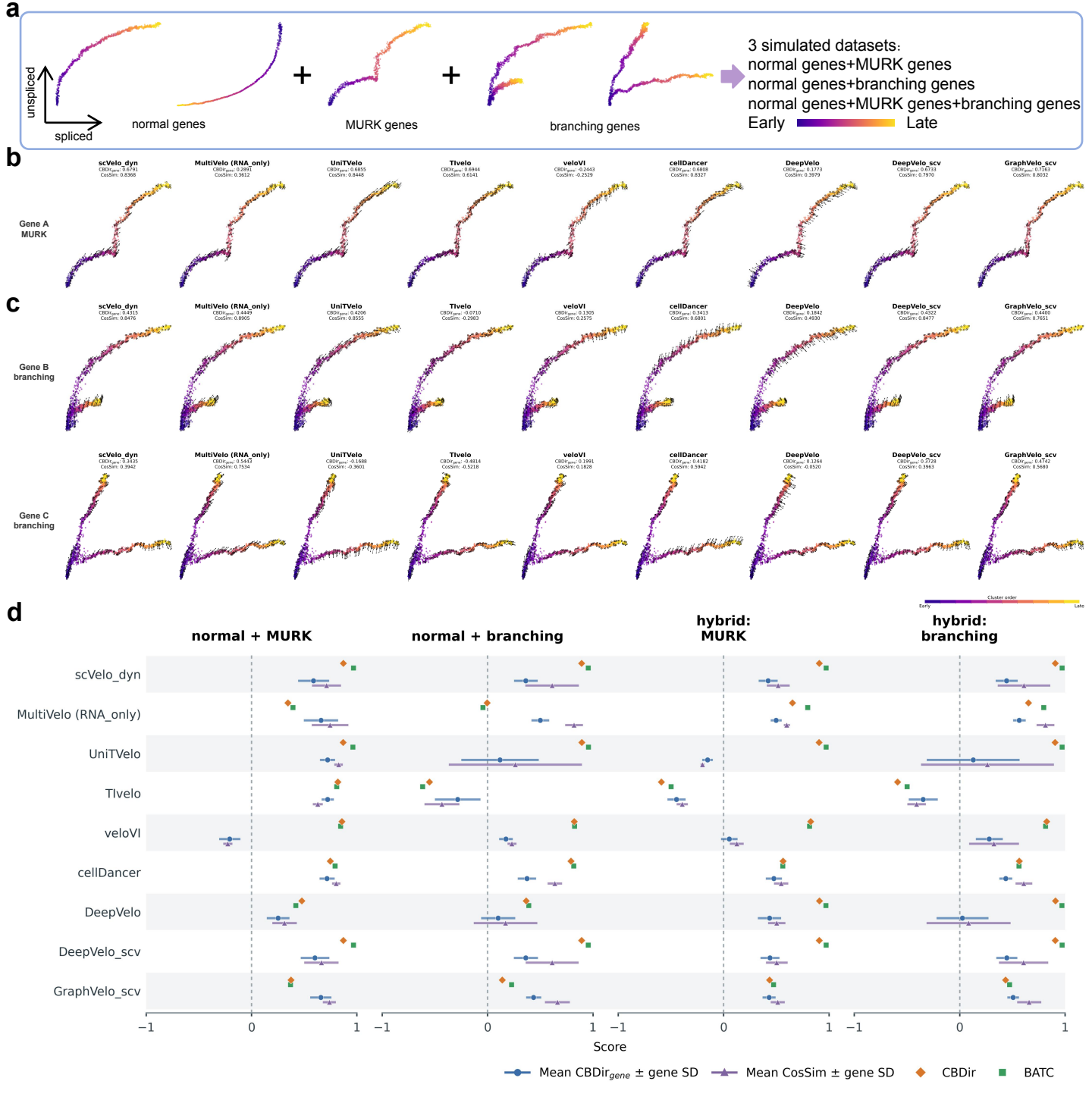

a
unspliced
spliced
normal genes
MURK genes
branching genes
3 simulated datasets:
normal genes+MURK genes
normal genes+branching genes
normal genes+MURK genes+branching genes
Early
Late
b
c
scVelo_dyn
MultiVelo (RNA_only)
UniTVelo
TIvelo
veloVI
cellDancer
DeepVelo
DeepVelo_scv
GraphVelo_scv
Gene A MURK
Gene B branching
Gene C branching
d
Cluster order
Early
Late
normal + MURK
normal + branching
hybrid: MURK
hybrid: branching
scVelo_dyn
MultiVelo (RNA_only)
UniTVelo
TIvelo
veloVI
cellDancer
DeepVelo
DeepVelo_scv
GraphVelo_scv
−1
0
1
Score
Mean CBDir$_{gene}$ ± gene SD
Mean CosSim ± gene SD
CBDir
BATC